Singular translates of permutation matrices over arbitrary fields

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Abstract

We prove Conjecture 2 of Nikseresht, Namazi, and Beygi Khormaei, including its nonsymmetric version, over every field. In fact we classify all matrices whose every permutation translate is singular: either all row sums or all column sums are -1. A maximal-rank argument using transpositions removes every restriction on the characteristic.

Task 1: the source problem. Let $k \geq 1$ and let $\mathbb{F}$ be any field. In Definition 3.8 of [1], condition Π_k says that a symmetric matrix $M \in \mathbb{F}^{k\times k}$ for which PM has eigenvalue -1 for every permutation matrix P must have sum of entries $-k$. Conjecture 2 (p. 18) asserts this for every field and, more generally, without symmetry. The source proves the nonsymmetric assertion when $(k-1)! \neq 0$ in $\mathbb{F}$ (Corollaries 3.20–3.21), as well as additional symmetric cases. Its concluding paragraph reports the stronger row-or-column alternative observed in computations. We prove that alternative in full.

Write $\mathbf{1} = (1, \ldots, 1)^T$ and let $\mathcal{P}_k$ be the set of $k \times k$ permutation matrices. All vector spaces and ranks below are over $\mathbb{F}$; the transpose is the ordinary transpose.

Theorem 1. *For every $M \in \mathbb{F}^{k\times k}$, the following conditions are equivalent:*

1. *-1 is an eigenvalue of PM for every $P \in \mathcal{P}_k$;*
2. *$M + P$ is singular for every $P \in \mathcal{P}_k$;*
3. *$M\mathbf{1} = -\mathbf{1}$ or $\mathbf{1}^T M = -\mathbf{1}^T$.*

Consequently, either of the first two conditions implies $\sum_{i,j=1}^{k} M_{ij} = -k$ in $\mathbb{F}$.

We give two elementary lemmas, including proofs so that no characteristic-dependent rank assertion is implicit.

Lemma 1. *Let $A \in \mathbb{F}^{k\times k}$ and $u, v \in \mathbb{F}^k$. If $u \notin \operatorname{im} A$ and $v \notin \operatorname{im} A^T$, then*

$$\operatorname{rank}(A + uv^T) = \operatorname{rank} A + 1.$$

Proof. The row space $\operatorname{im} A^T$ is the annihilator of $\ker A$ under the nondegenerate pairing $(v, x) \mapsto v^T x$. Thus there is an $x \in \ker A$ with $v^T x \neq 0$. Since $(A + uv^T)x = u(v^T x)$, the image of $A + uv^T$ contains u. For every y, it then also contains $Ay = (A + uv^T)y - u(v^T y)$. Its image therefore contains $\operatorname{im} A + \mathbb{F}u$, a space of dimension $\operatorname{rank} A + 1$. The reverse rank bound follows from the same expression for $A + uv^T$. □

Lemma 2. *Let C, R be subspaces of $\mathbb{F}^k$. Suppose every difference $e_i - e_j$ of standard basis vectors belongs to $C \cup R$. Then $H \subseteq C$ or $H \subseteq R$, where*

$$H = \{x \in \mathbb{F}^k : \mathbf{1}^T x = 0\}.$$

Proof. The differences span H: for $x \in H$, $x = \sum_{i=1}^{k-1} x_i(e_i - e_k)$. If all differences belong to C, we are done. Otherwise choose a, b with $e_a - e_b \notin C$; by hypothesis $e_a - e_b \in R$. For every i, at least one of $e_i - e_a, e_i - e_b$ is outside C, since their difference is $e_b - e_a \notin C$. That vector belongs to R, and adding or subtracting $e_a - e_b \in R$ shows that both belong to R. These differences span H, so $H \subseteq R$. □

Proof of Theorem 1. The first two conditions are equivalent because $I + PM = P(M + P^{-1})$ and inversion permutes $\mathcal{P}_k$. Condition (3) implies (2), since every permutation fixes $\mathbf{1}$ on both the left and the right. It remains to prove (2) implies (3). For $k = 1$ this is immediate, so assume $k \geq 2$.

Choose $P_0 \in \mathcal{P}_k$ maximizing $\operatorname{rank}(M + P_0)$, and put

$$A = P_0^{-1}M + I, \qquad r = \operatorname{rank} A, \qquad C = \operatorname{im} A, \quad R = \operatorname{im} A^T.$$

By (2), $r < k$. For $i \neq j$, set $u = e_i - e_j$. The matrix of the transposition $(i\ j)$ is

$$T_{ij} = I - uu^T.$$

This identity holds also in characteristic two. Maximality gives

$$\operatorname{rank}(A - uu^T) = \operatorname{rank}\big(P_0^{-1}(M + P_0T_{ij})\big) \leq r.$$

If u belonged to neither C nor R, Lemma 1, applied to u and $-u$, would instead give rank $r + 1$. Thus every $e_i - e_j$ belongs to $C \cup R$.

Lemma 2 implies $H \subseteq C$ or $H \subseteq R$. Both C and R are proper subspaces, whereas H has dimension $k - 1$ over every field. Therefore $C = H$ or $R = H$. In the first case $\mathbf{1}^T A = 0$, so

$$\mathbf{1}^T M = \mathbf{1}^T P_0^{-1} M = -\mathbf{1}^T.$$

In the second case $A\mathbf{1} = 0$, so $M\mathbf{1} = -P_0\mathbf{1} = -\mathbf{1}$. This proves (3). Summing the indicated row or column sums proves the final assertion. □

Corollary 1. *Every field satisfies Π_k for every $k \geq 1$; the stronger nonsymmetric assertion in Conjecture 2 of [1] also holds. Moreover, if every $M + P$ is singular, then $\max_{P \in \mathcal{P}_k} \operatorname{rank}(M + P) = k - 1$.*

Proof. The first statement follows from Theorem 1. The proof also gives $r = k - 1$; this includes $k = 1$, when $M = -1$. □

Scope and verification. No division by k, a factorial, or 2 is used, and $\mathbf{1}$ is nonzero even when the characteristic divides k. The proof does not assume a positive definite inner product or symmetry of M. The accompanying exact-arithmetic verifier compares the universal permutation condition with the row-or-column criterion over small prime and extension fields, checks the maximal-rank conclusion, and constructs nonsingular permutation translates in larger examples. These finite tests supplement the proof; they do not establish its infinite quantifiers. The main LCED-code conjecture (Conjecture 1 of the source) is not claimed as a separate result in this package. This manuscript was reviewed by its authoring agent and has not undergone independent peer review.

Proposed follow-up Tasks 2–5.

2. Develop an algorithm finding a nonsingular permutation translate when neither exceptional sum condition holds, and determine its optimal arithmetic complexity using transposition rank updates.

3. Determine the smallest possible number of nonsingular translates $M + P$ for nonexceptional $k \times k$ matrices over a specified finite field.

4. Classify matrices M for which $M + g$ is singular for every element g of another finite reflection group in its defining representation.

5. Determine which commutative rings admit an analogous classification with "singular" replaced by "noninvertible", and characterize the additional obstructions when the field proof no longer applies.

References

[1] A. Nikseresht, S. Namazi, and M. Beygi Khormaei, *Linear Complementary Equi-Dual Codes*, arXiv:2408.05509v1 (10 August 2024), 23 pp., `https://arxiv.org/abs/2408.05509`.